Postharvest Biology and Technology 211 (2024) 112843

Contents lists available at ScienceDirect

Postharvest Biology and Technology

journal homepage: www.elsevier.com/locate/postharvbio

Postharvest treatment with *Bacillus velezensis* LX mitigates disease incidence and alters the microbiome on kiwifruit surface

Zhihao duan[a,1], Huiyun Song[a,1], Honghui Shi[a,b], Zhu Gao[a,b,c], Jipeng Mao[a,b], Yunpeng Cao[d,**], Heqiang Huo[e], Juncheng Li[f,g,h], Xiaoling Wang[a,b,*], Mengfei Lin[a,b,*]

[a] Jiangxi Provincial Key Laboratory of Plantation and High Valued Utilization of Specialty Fruit Tree and Tea, Institute of Biological Resources, Jiangxi Academy of Sciences, Nanchang, Jiangxi, China
[b] Jiangxi kiwifruit engineering research center, Nanchang, Jiangxi, China
[c] Jinggangshan Institute of Biotechnology, Jiangxi Academy of Sciences, Ji'an, Jiangxi, China
[d] CAS Key Laboratory of Plant Germplasm Enhancement and Specialty Agriculture, Wuhan Botanical Garden, Chinese Academy of Sciences, Wuhan, China
[e] Mid-Florida Research & Education Center, IFAS, University of Florida, Apopka, FL 32703, USA
[f] Institute of Fruit Tree Research, Guangdong Academy of Agricultural Sciences, China
[g] Key Laboratory of South Subtropical Fruit Biology and Genetic Resource Utilization, Ministry of Agriculture and Rural Affairs, China
[h] Guangdong Provincial Key Laboratory of Tropical and Subtropical Fruit Tree Research, Guangzhou, China

ARTICLE INFO

Keywords:
Preservation
Actinidia chinensis
Microorganisms
Biological control

ABSTRACT

Microbial communities on plant surfaces are crucial in regulating plant growth and disease control. Exploring the composition of microorganisms on plant surfaces is essential for understanding their potential functions. The restructuring of microbial communities can help reduce postharvest diseases and maintain fruit quality. In this study, the use of *Bacillus velezensis* LX fermentation broth resulted in a reduction of *Diaporthe* and *Fusarium* abundance on the surface of kiwifruit, as well as changed in the fungal and bacterial communities on the kiwifruit surface, as determined by high-throughput sequencing. *B.velezensis* LX treatment reduces the natural decay incidence of postharvest kiwifruit but does not decrease the quality of kiwifruit. An analysis of the relationship between microbial taxa and fruit decay incidence revealed that several genera, including *Plectosphaerella*, *Vishniacozyma*, *Cryptococcus*, *Papiliotrema*, *Aureobasidium*, *Filobasidium*, *Actinomycetospora*, and *Parasutterella*, were enriched in the CK and LX treatment group during storage. This study contributes to a better understanding of how microbial antagonists regulate postharvest diseases in kiwifruit and provides insights for establishing a beneficial microbial synthetic flora to enhance the postharvest safety of kiwifruit.

1. Introduction

Kiwifruit (*Actinidia chinensis*) is a king of fruit, rich in nutrients such as vitamin C, amino acids, and minerals (Yang et al., 2024; Lin et al., 2023). Kiwifruit has many benefits, such as lowering blood pressure and preventing cancer, which are deeply loved by people (Lippi and Mattiuzzi, 2020). In recent years, kiwifruit has rapidly developed in domestic and foreign markets and has become a high-end fruit (Shu et al., 2023). However, kiwifruit is also a post-ripening fruit, and its fruit needs a period after harvesting to reach the unique flavor of the variety (Choi et al., 2022). During this process, kiwifruit is susceptible to fungal infections, which can cause the fruit to mold and deteriorate (Liu et al., 2018). According to surveys, the pathogens that cause kiwifruit rot include *Lasiodiplodia theobromae*, *Neofusicoccum parvum*, *Botryosphaeria dothidea*, *Diaporthe spiculosa*, *Fusarium fujikuroi*, etc., which can cause the mold rate of the fruit to be as high as 20%, causing severe economic losses (Dai et al., 2022). Therefore, effective anti-corrosion and preservation measures should be taken immediately after picking.

With the impact of global climate change, people are paying more and more attention to sustainable agriculture and food safety issues (Ren et al., 2020). In this context, the use of antagonistic microorganisms for the biological control of fungal diseases is becoming an increasingly

* Corresponding authors at: Jiangxi Provincial Key Laboratory of Plantation and High Valued Utilization of Specialty Fruit Tree and Tea, Institute of Biological Resources, Jiangxi Academy of Sciences, Nanchang, Jiangxi, China.
** Corresponding author.
E-mail addresses: xfcypeng@126.com (Y. Cao), WangXL_JXAS@163.com (X. Wang), linmengfeilixi@163.com (M. Lin).
[1] These authors have contributed equally to this work.

https://doi.org/10.1016/j.postharvbio.2024.112843
Received 14 January 2024; Received in revised form 8 February 2024; Accepted 9 February 2024
Available online 22 February 2024
0925-5214/

environmentally friendly, safe, economical, and effective alternative (Thambugala et al., 2020). The application of antagonistic microorganisms in the control of post-harvest diseases has the advantages of no toxic residues, environmental friendliness, safe application methods, effortless delivery, and economic production (Dukare et al., 2019). Many microorganisms have been reported to possess biological control abilities against pathogens. In a study conducted by Zhao et al. (2023), it was discovered that *Wickerhamomyces anomalus* is effective in controlling postharvest disease in kiwifruit. Similarly, Liu et al. (2020) reported that *Pichia kudriavzevii* can influence the fungal community succession of cherry tomatoes during refrigeration. The postharvest application of the yeast biocontrol agent *Metschnikowia fricicola* on apple and strawberry fruit significantly impacted microbial communities on fruit surface, identifying several potential beneficial genera for suppressing post-harvest diseases (Zhimo et al., 2021; Biasi et al., 2021). Additional studies have shown that treatment with *Debaryomyces hansenii* can inhibit plant pathogens, thereby reducing the occurrence of strawberry postharvest diseases and increasing ascorbic acid content (Zhao et al., 2023). Antagonistic microorganisms exhibit several modes of antibacterial activity, such as competition for nutrients and space, fungal parasitism, secretion of antifungal antibiotics and volatile metabolites, and induction of host resistance (Mendes et al., 2013). Antagonistic microorganisms mainly exist naturally in places unique to the surface of fruit and vegetables. In addition, microorganisms can also be obtained from other closely related or unrelated sources, such as phyllo spheres, roots, soil, and unique natural habitats (Wang et al., 2022).

Research in the past has mainly focused on studying the mechanism of action of antagonistic microorganisms, pathogens, and fruit (Shi et al., 2022a). However, an essential part of the microorganisms acting on the surface of the fruit has been ignored. Studies have shown that the microbial community on the surface of the fruit affects the occurrence of fruit rot, but it is not just the pathogen that acts alone (Wicaksono et al., 2023). The structure of the microbial community on the surface of fruit is crucial for the post-harvest storage process of the fruit. Preservation measures such as packaging treatment (Lin et al., 2019), hot chemical wet treatment (Chen et al., 2015), and waxing treatment (Abdelfattah et al., 2020) all significantly affect the resident microbial community of citrus fruit. Therefore, a better comprehension of the changes that happen on the surface of harvested fruit can lead to innovative post-harvest control methods. However, there is limited information available on the impact of antagonistic microbial agents on the microbial community on the surface of fruit.

To investigate the potential microbial antagonistic process on kiwifruit surfaces, aimed at inhibiting the growth of harmful microorganisms and ensuring fruit quality and yield, we screened for antagonistic microorganisms present on the kiwifruit surface. This study screened a strain of bacteria named LX from traditional pickled bamboo shoots. First, the hormone content in the fermentation broth of strain LX was determined, and the classification and gene function of strain LX were analyzed by analyzing its whole genome. Then, in a post-harvest preservation experiment of kiwifruit, strain LX was used to observe the effect of its fermentation broth on the growth of post-harvest pathogens of kiwifruit, the incidence rate during storage, and the quality of kiwifruit. By sequencing the amplified fragments of the conserved regions of 16 S and ITS in rDNA, we studied the changes in the bacterial and fungal communities on the surface of kiwifruit.

2. Materials and methods

2.1. Effect of LX fermented liquid on mycelial growth

5 μL of the overnight bacterial culture in LB medium (1.0×10^8 CFU/mL) were spotted in four directions on the PDA plate and inoculated with 7-day-old mycelia disks of *L. theobromae*, *N. parvum*, *B. dothidea*, *D. spiculosa*, or *F. fujikuroi* (5 mm diameter) at the center. Sterile saline served as a negative control in this test. The plates were incubated at 28 °C, and the radial growth of fungal colonies was measured daily for 7 d.

2.2. Effect of LX fermented liquid in controlling kiwifruit postharvest disease and regulating fruit quality

Kiwifruit (*Actinidia chinensis* cv. 'Jinyan') fruit were harvested from Fengxin County Doctor Kiwifruit Base (E114°45', N28°34'), Yichun City, Jiangxi Province, China, and promptly transported to the laboratory. Four hundred uniform fruit without physical injury and pest infestation were divided into two groups for soaking with sterile water (marked as control) and LX fermented liquid (1.0×10^8 CFU/mL, marked as treatment) for 10 minutes. After natural air drying, each fruit was transferred to a sterile plastic bag. The fruit were stored at 20 °C for 3, 6, 9, 12, and 15 d. Of the 200 fruit, 30 were labeled for disease index (DI) observation, and the remaining were designated for sampling. Three duplicates were performed for each treatment.

Fruit firmness was assessed by a GY-4 fruit firmness tester. Kiwifruit' soluble solid concentration (SSC), expressed in terms of Brix, was measured with a digital hand-held refractometer. The assay of titratable acid concentration was determined following the standard method. Phytohormones in the LX fermentation broth that includes gibberellin (GA), jasmonic acid (JA), Indole-3-butyric acid (IBA), abscisic acid (ABA), were extracted with acetonitrile, concentrated, and then dissolved in a methanol solution. Finally, UPLC-ESI-MS/MS system was used to analyze the sample extracts.

2.3. LX genome sequencing and phylogenetic analysis

The genomic DNA of LX was extracted using the Hipure bacterial DNA kit. The whole genome was sequenced at BMKCloud (www.biocloud.net) using both the Pacific Biosciences platform and the Illumina Miseq platform. The filtered subreads underwent assembly with Canu v1.5 software, followed by circularization using Circlator v1.5.5.

The whole genome sequences of 12 strains were obtained from NCBI with the following accession numbers: CP069214.1, CP126226.1, NZ_CP041361.1, NZ_CP043416.1, NZ_CP043546.1, NZ_CP045711.1, NZ_CP047157.1, NZ_CP051011.1, NZ_CP051463.1, NZ_CP053376.1, NZ_CP053377.1, NZ_CP054467.1. A multilocus sequence analysis (MLSA) employing six housekeeping genes (16 S rRNA, dnaA, gyrB, mutL, recA, and rpoB) was conducted to construct the phylogenetic tree. Sequence alignment was performed using MAFFT, sequence trimming was done using MEGA, and splicing and construction of housekeeping genes were done using raxmlGUI 2.0.

2.4. High-throughput 16 S ribosomal RNA and ITS gene sequencing

TGuide S96 Magnetic Soil/Stool DNA Kit was used to extract kiwifruit samples total genomic DNA. Primer pairs 338 F: 5′- ACTCCTACGGGAGGCAGCA-3′ and 806 R: 5′- GGACTACHVGGGTWTCTAAT-3′ were used to amplify bacterial 16 S rRNA and primer pairs F: 5′-CTTGGTCATTTAGAGGAAGTAA-3′ and R: 5′-GCTGCGTTCTTCATCGATGC-3′ were used to amplify ITS genes. PCR products of 16 S rRNA and ITS were checked on agarose gel and purified through the Omega DNA purification kit. Then the paired ends (2×250 bp) were performed on the Illumina Novaseq 6000 platform.

2.5. Bioinformatic analysis

USEARCH (version 10.0) was used to allocate qualified sequences with more than 97% similarity thresholds to one operational taxonomic unit (OTU). Naive Bayes classifier in QIIME2 (Bolyen et al., 2019) and SILVA database (Quast et al., 2012) with a confidence threshold of 70% were used to perform the Taxonomy annotation of the OTUs/ASVs. Alpha diversity was performed to identify each sample's species diversity complexity utilizing QIIME2 software. Principal coordinate analysis (PCoA) and permutational multivariate analysis of variance

(PERMANOVA) were conducted to identify variations in fungal and bacterial composition between fruit treated with LX and those untreated using Bray-Curtis dissimilarity. Two-way analysis of variance was employed to compare bacterial and fungal abundance and diversity by GraphPad prism 8. Linear discriminant analysis (LDA) coupled with effect size (LefSe) was employed to assess taxa that exhibited differential abundance. The relationships between natural decay incidence, weight loss, firmness, titratable acids, and total soluble solids with the top 20 genera of bacterial and fungal communities are depicted using heat maps. Gephi software was utilized to generate a network illustrating the relationships among predominant genera. Alpha and beta diversity, Spearman correlation, and Lefse analyses were performed at the BMKCloud (www.biocloud.net).

3. Results

3.1. LX has been identified as a *Bacillus velezensis*

To determine the phylogenetic status of LX, we sequenced its entire genome, which is 3737,142 bp in length (Fig. 1A). Using multilocus sequence analysis (MLSA) with six housekeeping genes (16 S rRNA, dnaA, gyrB, mutL, recA, and rpoB), we inferred a tree that showed LX on a separate branch, with *Bacillus velezensis* GUAL210 as its closest neighbor (Fig. 1B).

3.2. LX can inhibit post-harvest pathogens in kiwifruit

LX colony is milky white, circular, with a smooth and slightly convex surface, 1.0–1.5 mm, and a sticky texture (Fig. 2A). The inhibitory effect of LX bacterial strain fermentation broth on common post-harvest pathogens of kiwifruit was measured. The results showed that the fermentation broth of the LX strain could inhibit the growth of all pathogens (Fig. 2C). Among them, the growth of *D. spinulose* (1.75 cm), *F. fujikuroi* (1.47 cm), and *B. dothidea* (1.23 cm) was strongly inhibited, with an inhibition diameter greater than 1.2 cm. The growth inhibition of *L. theobromae* (0.51 cm) and *N. parvum* (0.87 cm) was weaker, with an inhibition diameter of less than 1 cm (Fig. 2B; Table S1).

3.3. The effect of preharvest application of LX on postharvest disease and kiwifruit quality

The effects of LX treatment on the natural decay incidence and fundamental quality indicators of kiwifruit are shown in Fig. 3. The results indicate that LX treatment reduces the natural decay incidence of kiwifruit (Fig. 3A, B), especially during the critical period of kiwifruit preservation. At the same time, LX treatment increases the firmness of kiwifruit, effectively extending the time of softening and decay after kiwifruit harvesting (Fig. 3D). The treatment of LX has no effect on weight loss of kiwifruit (Fig. 3C). The quality of kiwifruit is mainly affected by the content of titratable acids and total soluble solids during kiwifruit storage. The titratable acid content was lower than that of the control group at 6 and 9 d, with no difference observed at other time points (Fig. 3E). The treatment of LX has no effect on the content of total soluble solids (Fig. 3F). Therefore, LX treatment will not reduce the quality of kiwifruit.

3.4. Overview of bacteria and fungi on the surface of kiwifruit

The experiment results show that the number of bacterial (Table S2) OTUs on the kiwifruit epidermis in the CK group was 13,985, of which the OTUs of CK1D, CK5D, CK7D, CK9D, and CK11D were 2831, 4121, 2530, 2924, and 1579, respectively. The number of bacterial OTUs on the kiwifruit epidermis in the LX treatment group was 11,496. Among them, the OTUs of LX1D, LX5D, LX7D, LX9D, and LX11D were 2008, 2312, 2888, 2532, and 1756, respectively. The number of fungal (Table S2) OTUs in the CK group was 3966, markedly lower compared to the bacterial count. The fungal OTUs of CK1D, CK5D, CK7D, CK9D, and CK11D were 1089, 991, 783, 648, and 455, respectively. The number of

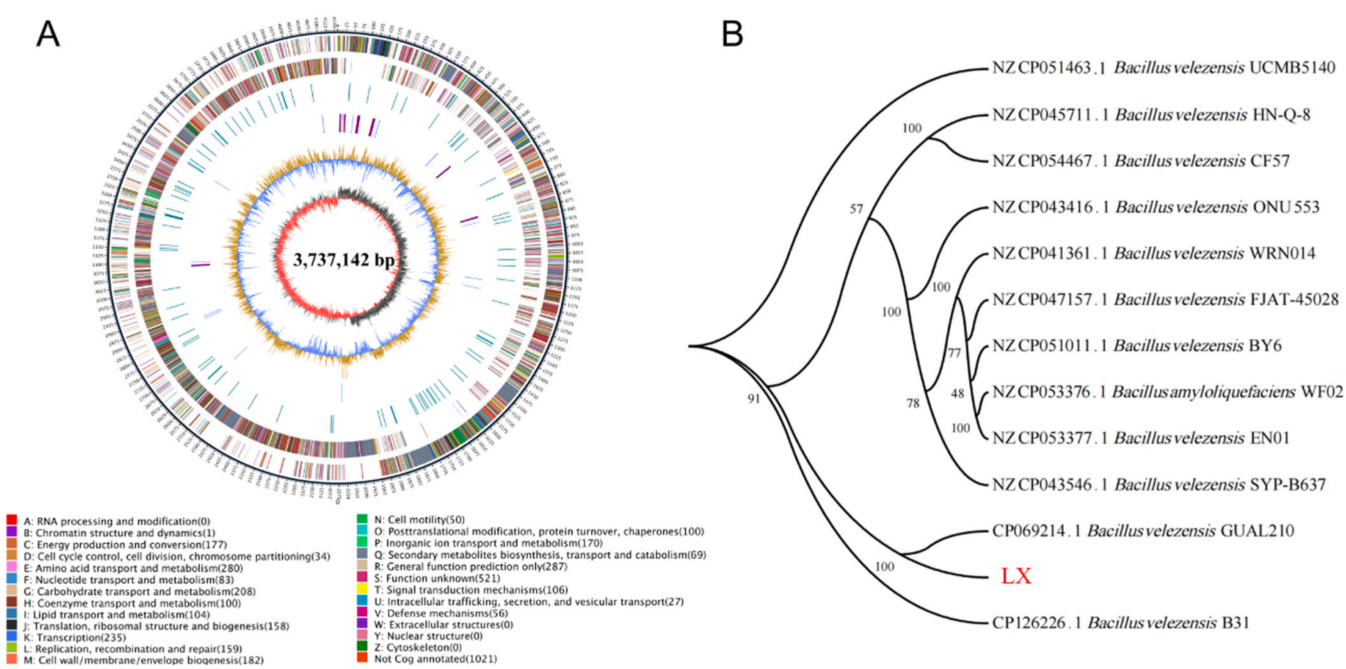

Fig. 1. Genome circle of LX (A) and phylogenetic analysis tree of LX and other bacteria (B). (A) The outermost circle of the figure indicates the genome size, with each scale representing 5 kb. The second and third circles depict genes on the positive and negative strands of the genome, respectively, with different colors denoting various COG functional classifications. The fourth circle represents repetitive sequences, while the fifth circle illustrates tRNA and rRNA, where blue indicates tRNA and purple represents rRNA. The sixth circle displays GC content, where light yellow signifies higher GC content than the average genome, and blue indicates lower GC content. The innermost circle depicts GC-skew, where dark gray represents regions with higher G content than C, and red represents regions with higher C content than G. (B) Tree diagram based on 6 housekeeping genes. Maximum likelihood-based phylogenetic inference was conducted using raxmlGUI. Bootstrap values for 500 replicates indicate the significance of each branch, and the numbers at the branches represent confidence values of clustered taxa in the tree.

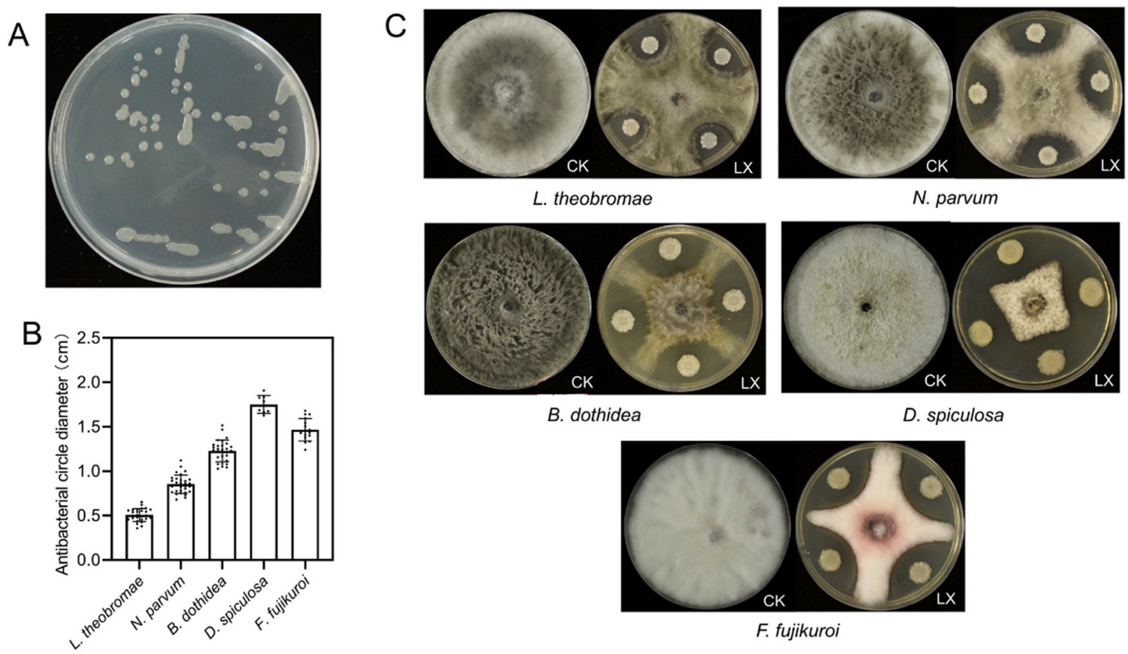

Fig. 2. LX morphology on LB plates (A) and the statistical analysis (B) and images (C) of LX's inhibitory effect on different postharvest kiwifruit pathogenic bacteria.

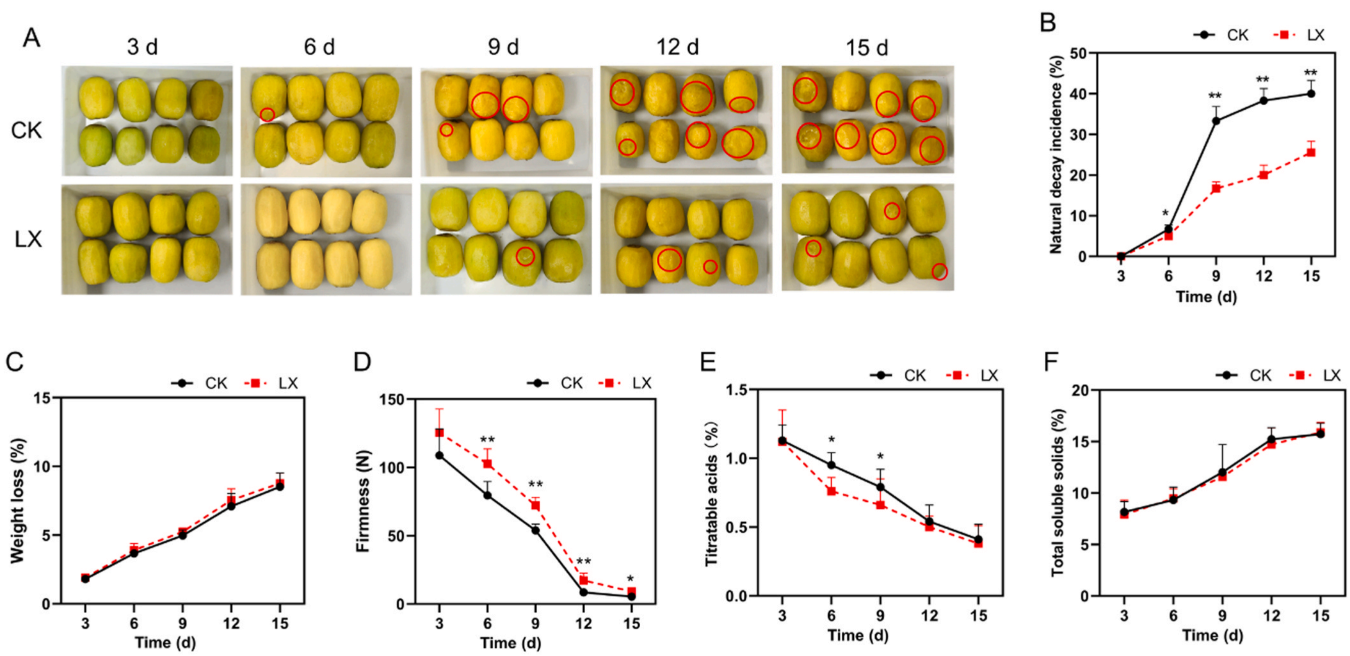

Fig. 3. Picture of kiwifruit peeling (A) and analysis of Natural decay incidence (B), weight loss (C), firmness (D), titratable acid (E) and Total ascorbic acid (F) of kiwifruit over fifteen days of storage after preharvest application of LX. * $P < 0.05$, ** $P < 0.001$.

fungal OTUs treated with LX on the kiwifruit surface was 2404. The OTUs of LX1D, LX5D, LX7D, LX9D, and LX11D were 531, 211, 1015, 334, and 313, respectively. The LX treatment reduced the bacteria and fungi on the surface of kiwifruit.

3.5. α and β diversity on kiwifruit surface

The bacterial and fungal diversity of kiwifruit skin was evaluated using the Shannon diversity index (Fig. 4). The alpha diversity of bacteria and fungi in response to LX application was influenced by both processing and storage time, as well as the bidirectional interaction between processing and storage time (ANOVA, $P \leq 0.05$) (Table 1). The bacterial alpha diversity of LX-treated and untreated fruit differed significantly. There was a distinction in bacterial alpha diversity between LX-treated and untreated fruit. (Fig. 4A, C).

Based on paired comparisons of storage time, the bacteria diversity of LX-treated fruits was lower compared to the control fruit at 3, 6 and 12 d. However, the diversity disparity between the two groups at 9 and 15 d did not demonstrate statistical significance (Fig. 4B). The overall fungal diversity of LX-treated and untreated fruit differed significantly. At 3, 6, 9 and 15 d, the diversity of treated fruit was lower than that of control fruit. The diversity disparity between the two groups at 12 d did not show statistical significance (Fig. 4D).

PCoA was employed for visualizing the Bray-Curtis differences

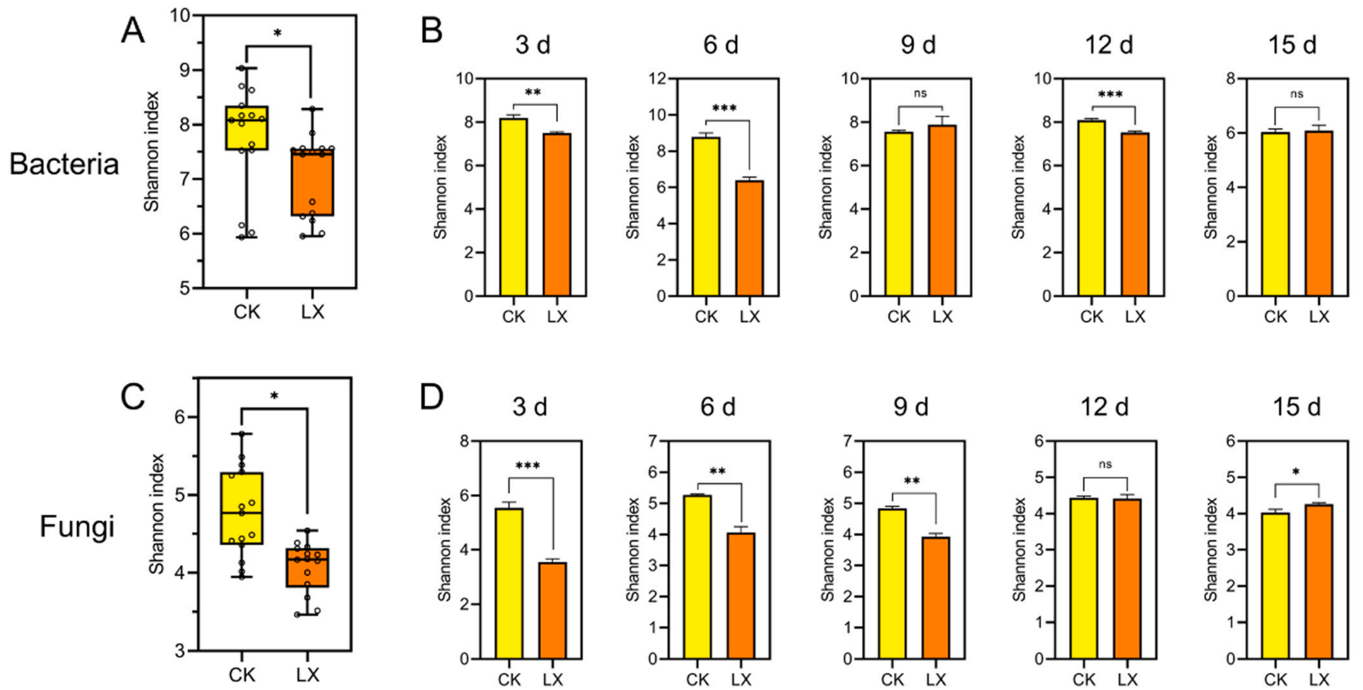

Fig. 4. Box plots of Shannon index (reflect α diversity) of the bacterial (A, B) and fungal (C, D) communities of kiwifruit samples. * $P < 0.05$, ** $P < 0.001$, *** $P < 0.0001$.

Table 1
The effects of treatment and storage time and their interaction on Shannon index of kiwifruit fruit (two-way ANOVA analysis).

	Bacteria		Fungi	
	F-value	*P*-value	F-value	*P*-value
Treatment	60.11	0.0162	312.8	0.0032
Storage time	79.91	<0.0001	25.98	0.0152
TreatmentxStorage time	119.8	<0.0001	126.5	0.0062

Table 2
The effects of treatment and storage time on beta diversity based on Bray–Curtis distance of kiwifruit fruit (PERMANOVA analysis).

	Bacteria		Fungi	
	R^2	*P*-value	R^2	*P*-value
Treatment	0.042	0.001	0.095	0.001
Storage time	0.164	0.001	0.177	0.005

between samples (Fig. 5), and PERMANOVA was conducted to verify variations in the structure of bacterial and fungal communities among distinct samples (Table 2). The PERMANOVA results indicated a significant impact of LX treatment on the bacterial and fungal community structure of kiwifruit ($P = 0.001$) The PERMANOVA results revealed an effect of LX treatment on the bacterial and fungal community structure of kiwifruit. Additionally, storage time emerged as a factor altering the fungal and bacterial community structure post-LX treatment.

3.6. Comparison of bacterial and fungal communities across various samples

KRONA species annotation was employed to assess the abundance of

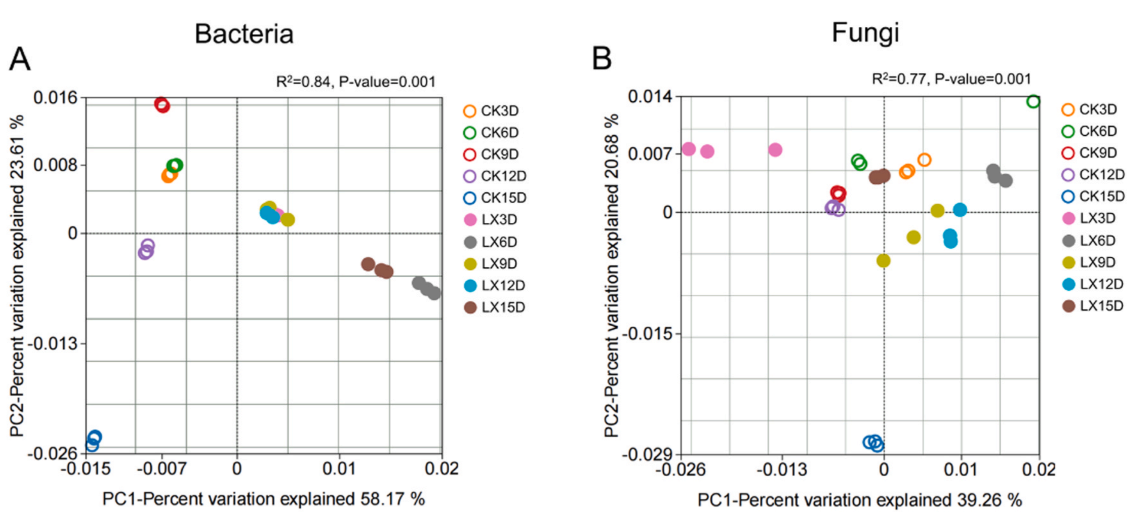

Fig. 5. The principal coordinate analysis (PCoA) of bacterial (A) and fungal (B) communities on kiwifruit surface based on Bray-Curtis dissimilarity during storage.

taxa (from phylum to order) of bacteria and fungi in both treated and untreated fruit (Table S3). The results showed that *Ascomycota* (LX: 57% and CK: 77%) and *Basidiomycota* (LX: 43% and CK: 23%) constituted the dominant phyla within the fungal community. At the class level, *Dothideomycetes* (LX: 52% and CK: 63%), *Exobasidiomycetes* (LX: 6% and CK: 9%), and *Tremellomycetes* (LX: 36% and CK: 7%) were the dominant groups. *Pleosporales* (LX: 26% and CK: 36%), *Cladosporiales* (LX: 22% and CK: 18%), and *Tremellales* (LX: 35% and CK: 6%) emerged as the dominant groups at the order level. In the bacterial community, *Proteobacteria* (LX: 40% and CK: 47%) was the dominant phylum. At the class level, *Gammaproteobacteria* (LX: 24% and CK: 25%), *Alphaproteobacteria* (LX: 16% and CK: 23%), *Bacteroidia* (LX: 13% and CK: 15%), and *Bacilli* (LX: 27% and CK: 4%) were the dominant groups. At the order level, *Bacillales* (LX: 23% and CK: 0.2%) and *Xanthomonadales* (LX: 9% and CK: 14%) were the dominant groups. At the genus level, the top 10 relative abundances of bacterial and fungal genera are presented (Fig. 6A, B; Table S4).

The LX-treated group exhibited a lower relative abundance of *Diaporthe* and *Fusarium*, the most prevalent post-harvest pathogens of kiwifruit, compared to the control group at 9, 12 and 15 d (Fig. 6C, D). This indicates that LX treatment can reduce the abundance of *Diaporthe* and *Fusarium* on the surface of kiwifruit in a short time, thereby achieving the purpose of preservation. This result aligns with the findings from the plate confrontation test. Other post-harvest pathogens of kiwifruit were not detected.

3.7. Correlations between the predominant genera and natural decay incidence of kiwifruit and the microbial co-occurrence networks in response to LX

Spearman correlation was employed to assess the correlation between the dominant genera (top 20 bacterial and fungal genera) in both the treatment and control groups. The analysis also considered natural decay incidence, weight loss, firmness, titratable acids, and total soluble solids, with a particular emphasis on their relationship with fruit rot incidence (Fig. 7). The findings indicated a positive correlation between unclassified-Bacteria in the CK group and fruit rot incidence, suggesting a potential role of these bacteria in promoting fruit rot within the control group (Fig. 7A). *Lachnospiraceae*, *Faecalibacterium*, *Muribaculaceae*, *Allorhizobium-Neorhizobium-Pararhizobium-Rhizobium*, *Novosphingobium*, *Bacteroides*, and *Terriglobus* were negatively correlated with fruit rot incidence. Within the treatment group, *Curtobacterium*, *Enterobacterales*, *Enterobacter,* and *Pantoea* exhibited positive correlations with fruit rot incidence. Meanwhile, *Sphingomonas*, *1174–901–12*, and *Methylobacterium-Methylorubrum* exhibited negative correlations with fruit rot incidence. Regarding fungal genera (Fig. 7B), *Diaporthe*, *Sporidiobolus*, *Cladosporium*, and *Zasmidium* in the CK group displayed positive correlations with rot incidence, while *Cyphellophora*, *Pleosporales*, and *unclassified-Fungi* were negatively correlated. *Cryptococcus*, *Papiliotrema*, *Aureobasidium*, and *Filobasidium* exhibited positive correlations with rot incidence in the treatment group, and no fungal genera demonstrated negative correlations with rot incidence.

Gephi software was employed to visualize the co-occurrence network in both groups throughout storage. Within the CK group (Fig. 8A), two genera, *Actinomycetospora* and *Parasutterella*, exhibited positive associations with *Diaporthe*. Simultaneously, two genera, *Allorhizobium_Neorhizobium_Pararhizobium_Rhizobium* and *unclassified_Enterobacteriaceae*, displayed negative associations with *Diaporthe*. In the LX treatment group (Fig. 8B), three genera, *Plectosphaerella*, *Cryptococcus*, and *unclassified_Enterobacterales*, demonstrated positive associations with *Fusarium*.

3.8. Lefse analysis

Lefse analysis, utilizing linear discriminant analysis (LDA), was employed to discern microbial changes between groups throughout the entire storage period. Different genera were found to be significantly different (LDA score > 4.0, $P < 0.05$), explaining the differences between LX-treated and untreated fruit. Five bacterial genera (Fig. 9A), *Bacillus*, *Sphingomonas*, *Pantoea*, *Stenotrophomonas*, and unclassified *Bacteria*, and eight fungal genera (Fig. 9B), *Aureobasidium*, *unclassified_Mycosphaerellaceae*, *Neosetophoma*, *Phaeosphaeria*, *Articulospora*, *Sporidiobolus*, *Hannaella*, and *Vishniacozyma*, were differentially enriched in stored fruit.

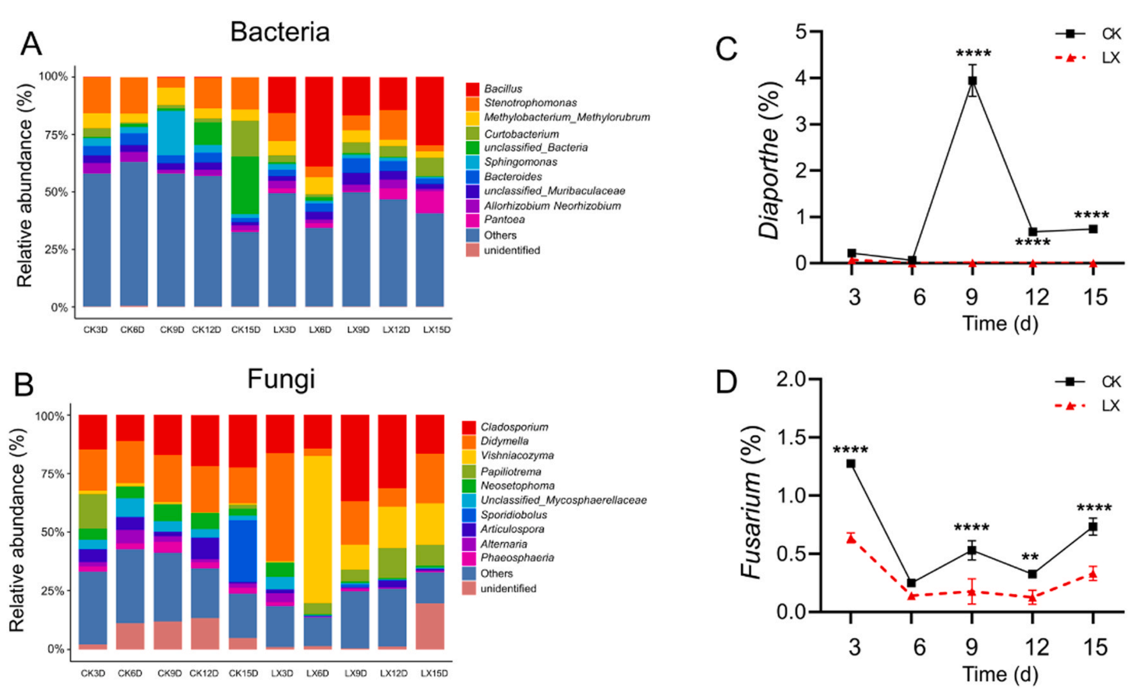

Fig. 6. Relative abundance of bacterial (A) and fungal (B) taxa at the genus rank and kiwifruit postharvest pathogenic bacteria (C, D) was shown. The relative abundance of the top 10 was given in detail in all samples (Table S4). ** $P < 0.001$, *** $P < 0.0001$, ****$P < 0.00001$.

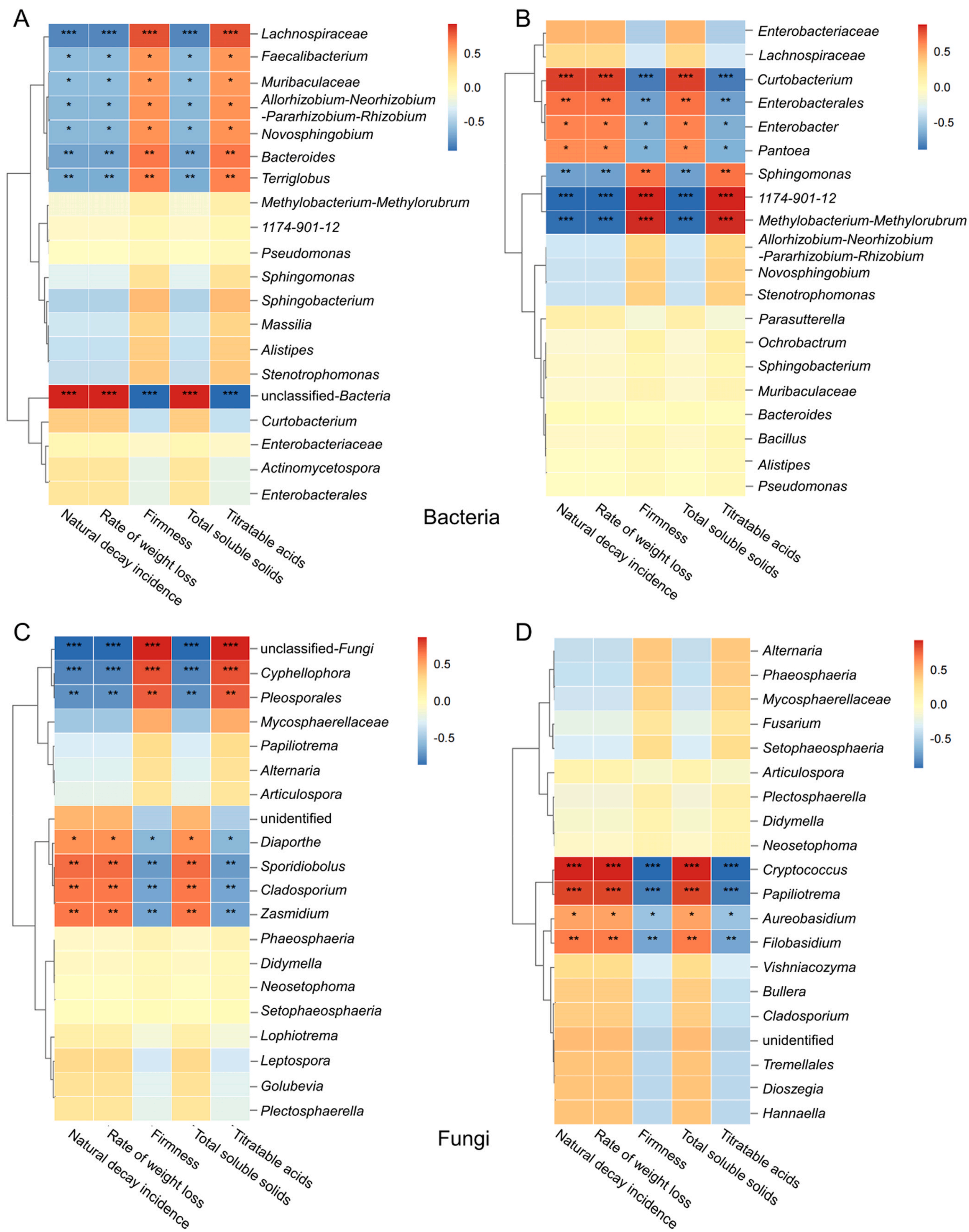

Fig. 7. Correlation (Spearman's) of the natural decay incidence, rate of weight loss, firmness, titratable acid and total ascorbic acid with kiwifruit bacterial and fungal genera. Red color represents a positive correlation and blue color represents a negative correlation. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

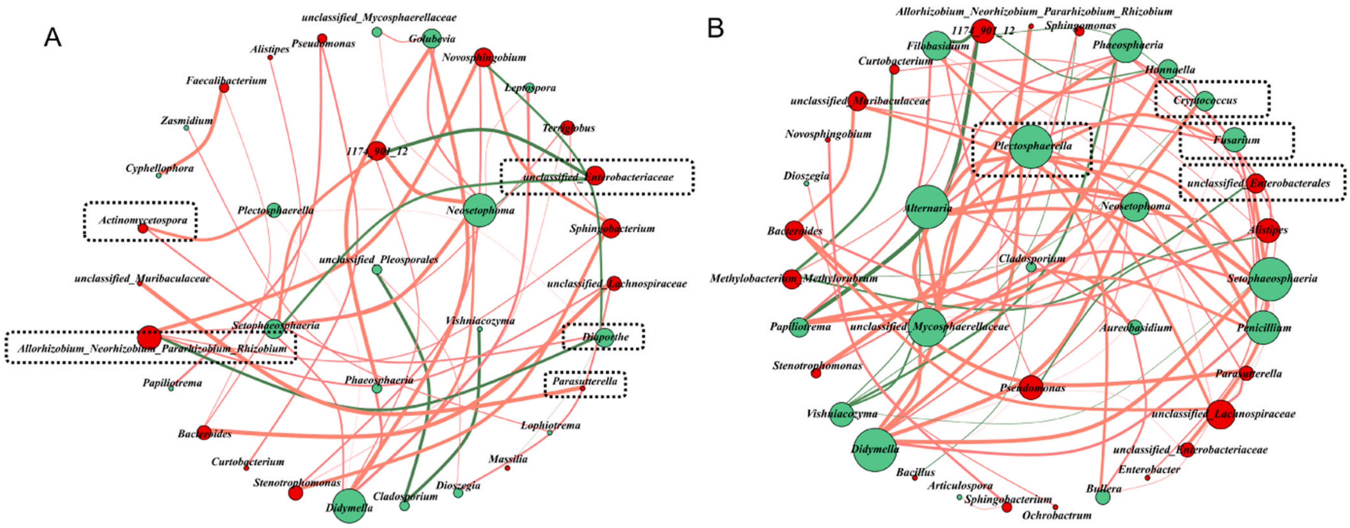

Fig. 8. Network co-occurrence analysis (Spearman $|\rho| > 0.7$ and $p < 0.001$) of bacterial and fungal communities in the CK (A) and LX treatment (B) groups during storage. Red lines represent a positive correlation and green lines represent a negative correlation. Line thickness represents the degree of the correlation. The green circle is bordered by fungi, the red circle represents bacteria, and the size of the circle represents the number of associated bacteria.

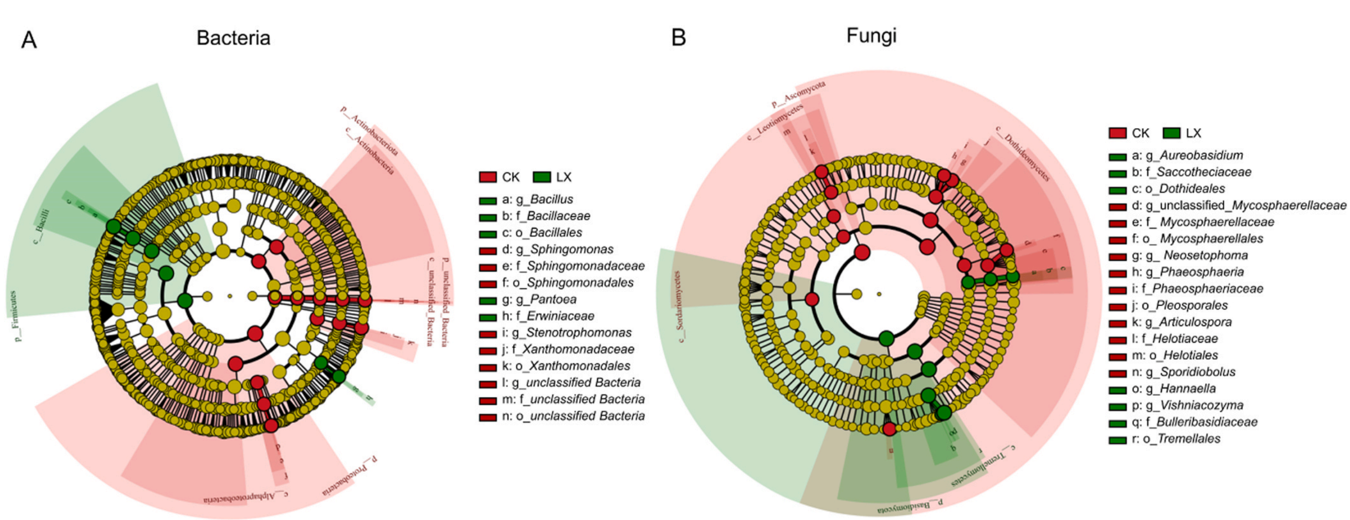

Fig. 9. Evolutionary branching diagram of bacteria (A) and fungi (B) based on Lefse analysis. The circles radiating from inside to outside in the evolutionary branch diagram represent the classification levels of phyla, class, order, family, genus, and species; Each small circle at different classification levels represents a classification at that level, and the diameter of the small circle is proportional to the relative abundance; The red and green circles represent species with significant differences, while the yellow circles represent species with significant differences.

4. Discussion

In our study, LX was identified through whole-genome sequencing as a *Bacillus velezensis* strain isolated from traditional sour bamboo shoots (Fig. 1). *B. velezensis* LX effectively inhibits the growth of postharvest pathogenic bacteria in kiwifruit, particularly *D. spiculosa* and *F. fujikuroi* (Fig. 2). In addition, the application of LX to postharvest kiwifruit reduced the richness of *Diaporthe* and *Fusarium* within 15 d (Fig. 6C, D). Previous studies have demonstrated that *B. velezensis* strains can stimulate plant growth, produce diverse bioactive secondary metabolites, and inhibit plant pathogenic bacteria (Adeniji et al., 2019). For instance, *B. velezensis* K-9 has successfully prevented potato scab and increased potato yield, potentially serving as a biocontrol agent for managing potato scab (Ma et al., 2023). Additionally, its resistance to gray mold has led to its formulation into a commercial fungicide against *Botrytis cinerea*, the causative agent of gray mold, which infects over 200 plant species worldwide (Rabbee et al., 2019). *B. velezensis* has also been effectively employed in postharvest preservation technology for citrus fruit, efficiently antagonizing the citrus causal agents *Colletotrichum gloeosporioides* and *Penicillium digitatum* in vitro and in vivo (Vu et al., 2023). However, the mechanism of postharvest fruit preservation by *B. velezensis* remains unclear. It is evident that the application of *B. velezensis* affects the microbial community on the surface of postharvest fruit, with a high abundance of microorganisms and microbial taxa reported to be beneficial to plants, potentially contributing to the preservation of these fruit (Shi et al., 2022b). Therefore, this study aims to analyze the effect of post-harvest application of *B. velezensis* LX on the surface microbiota of kiwifruit during storage to discover the microbial community structure beneficial to kiwifruit preservation.

The application of LX fermentation broth to kiwifruit after harvest reduced the natural decay rate and increased the fruit hardness (Fig. 3). This effect may be attributed to metabolites in the LX fermentation broth inhibiting the growth of kiwifruit rot pathogens. Additionally, various plant hormones in the LX fermentation broth, including SA (711.4 ng/mL), IAA (140.3 ng/mL), and JA (46.9 ng/mL) are speculated to play a crucial role (Table S5). Research has indicated that salicylic acid protects postharvest red jujube fruit from *Alternaria alternata* by activating defense-related systems rather than through fungicidal activity (Cao et al., 2013). Sun et al. (2023) discovered that inhibiting the crosstalk between JA and SA signaling pathways and the MAPK3 pathway can lead to the independent function of the BR pathway, increasing cherry tomato fruit resistance and reducing disease incidence by 66%.

Correlation analysis (Fig. 7) between dominant genera and natural decay incidence revealed that, for bacteria, *Curtobacterium* and *Enterobacterales* in the CK group showed no correlation with the fruit rot rate. However, they were positively correlated with the fruit rot rate after LX treatment. *Sphingomonas* was not correlated with the fruit rot rate in the CK group but was negatively correlated with the fruit rot rate after LX treatment. Similarly, *Cladosporium* was positively correlated with the fruit rot rate for fungi but showed no correlation after LX treatment. On the other hand, *Papiliotrema* displayed no correlation with the fruit rot rate in the CK group but was positively correlated with the fruit rot rate after LX treatment. These findings suggest that using LX may alter the function of specific taxa, and additional experiments are required to confirm this hypothesis.

The application of LX to kiwifruit after harvest altered the dominant microbial community on the fruit's surface. Regarding bacteria, the abundance of *Bacillus* in the CK group was only approximately 0.15%, but it became the dominant genus with the highest abundance after LX treatment (Table S4). This change is attributed to the fact that the sprayed LX belongs to the genus *Bacillus*, indirectly confirming the reliability of the test results. The abundance of *Vishniacozyma* in fungi also markedly increased after LX treatment. Moreover, LefSe analysis demonstrated significant enrichment of *Vishniacozyma* in the LX-treated group (Table S4). *Vishniacozyma* has been identified as a potential beneficial endophyte in kiwifruit, effectively inhibiting *Botrytis cinerea* in kiwifruit fruit (Nian et al., 2023). Additionally, *Vishniacozyma* has been utilized for the control of fungal diseases in organic pears, including *Penicillium expansum*, *Botrytis cinerea*, and *Cladosporium sp.* (Gorordo et al., 2022). Therefore, we hypothesize that *Vishniacozyma* constitutes a crucial component of the microorganisms on the surface of kiwifruit, contributing to its preservation.

5. Conclusion

In this study, we observed that *Bacillus velezensis* LX fermentation broth suppressed the growth of postharvest kiwifruit pathogens and induced significant alterations in the fungal and bacterial communities on the kiwifruit surface. These changes involved increases in fungal diversity and species-specific variations. Correlation analysis between dominant microbial genera and postharvest kiwifruit rot incidence revealed several potentially beneficial fungal and bacterial taxa. Combining beneficial microorganisms and biological control agents to create complex inoculants requires further in-depth research and exploration to comprehend the potential biocontrol mechanisms governing their collaborative action on the fruit surface. This investigation contributes to a better understanding of how microbial antagonists regulate postharvest diseases in kiwifruit and provides insights for establishing a beneficial microbial synthetic flora to enhance the postharvest safety of kiwifruit.

Funding

This work is supported by the Jiangxi Talent Team Plan [20232BCJ23045], Jiangxi Natural Science Foundation [20232BAB215039], Scientific Research Project of Jiangxi Academy of Sciences [2023YSBG50002, 2022YSBG22004, 2022YYB21, 2022YRCC001].

CRediT authorship contribution statement

Zhihao Duan: Writing – original draft, Data curation. **Mengfei Lin:** Project administration, Funding acquisition, Formal analysis, Conceptualization. **Xiaoling Wang:** Funding acquisition, Conceptualization. **Juncheng Li:** Software, Data curation. **Heqiang Huo:** Validation, Supervision. **Yunpeng Cao:** Supervision, Software, Methodology. **Jipeng Mao:** Project administration, Investigation, Formal analysis. **Zhu Gao:** Project administration, Formal analysis, Conceptualization. **Honghui Shi:** Resources, Investigation. **huiyun song:** Writing – original draft, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data required for the article is provided in pictures and tables.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.postharvbio.2024.112843.

References

Abdelfattah, A., Whitehead, S.R., Macarisin, D., Liu, J., Burchard, E., Freilich, S., Dardick, C., Droby, S., Wisniewski, M., 2020. Effect of washing, waxing and low-temperature storage on the postharvest microbiome of apple. Microorganisms 8 (6), 944. https://doi.org/10.3390/microorganisms8060944.

Adeniji, A.A., Loots, D.T., Babalola, O.O., 2019. *Bacillus velezensis*: phylogeny, useful applications, and avenues for exploitation. Appl. Microbiol. Biotechnol. 103, 3669–3682. https://doi.org/10.1007/s00253-019-09710-5.

Biasi, A., Zhimo, V.Y., Kumar, A., Abdelfattah, A., Salim, S., Feygenberg, O., Wisniewski, M., Droby, S., 2021. Changes in the fungal community assembly of apple fruit following postharvest application of the yeast biocontrol agent *Metschnikowia fructicola*. Horticulturae 7 (10), 360. https://doi.org/10.3390/horticulturae7100360.

Bolyen, E., Rideout, J.R., Dillon, M.R., Bokulich, N.A., Abnet, C.C., Al-Ghalith, G.A., Alexander, H., Alm, E.J., Arumugam, M., Asnicar, F., Bai, Y., Bisanz, J.E., Bittinger, K., Brejnrod, A., Brislawn, C.J., Brown, C.T., Callahan, B.J., Caraballo-Rodríguez, A.M., Chase, J., Cope, E.K., Da Silva, R., Diener, C., Dorrestein, P.C., Douglas, G.M., Durall, D.M., Duvallet, C., Edwardson, C.F., Ernst, M., Estaki, M., Fouquier, J., Gauglitz, J.M., Gibbons, S.M., Gibson, D.L., Gonzalez, A., Gorlick, K., Guo, J., Hillmann, B., Holmes, S., Holste, H., Huttenhower, C., Huttley, G.A., Janssen, S., Jarmusch, A.K., Jiang, L., Kaehler, B.D., Kang, K.B., Keefe, C.R., Keim, P., Kelley, S.T., Knights, D., Koester, I., Kosciolek, T., Kreps, J., Langille, M.G.I., Lee, J., Ley, R., Liu, Y.-X., Loftfield, E., Lozupone, C., Maher, M., Marotz, C., Martin, B.D., McDonald, D., McIver, L.J., Melnik, A.V., Metcalf, J.L., Morgan, S.C., Morton, J.T., Naimey, A.T., Navas-Molina, J.A., Nothias, L.F., Orchanian, S.B., Pearson, T., Peoples, S.L., Petras, D., Preuss, M.L., Pruesse, E., Rasmussen, L.B., Rivers, A., Robeson, M.S., Rosenthal, P., Segata, N., Shaffer, M., Shiffer, A., Sinha, R., Song, S.J., Spear, J.R., Swafford, A.D., Thompson, L.R., Torres, P.J., Trinh, P., Tripathi, A., Turnbaugh, P.J., Ul-Hasan, S., van der Hooft, J.J.J., Vargas, F., Vázquez-Baeza, Y., Vogtmann, E., von Hippel, M., Walters, W., Wan, Y., Wang, M., Warren, J., Weber, K. C., Williamson, C.H.D., Willis, A.D., Xu, Z.Z., Zaneveld, J.R., Zhang, Y., Zhu, Q., Knight, R., Caporaso, J.G., 2019. Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. Nat. Biotechnol. 37, 852–857. https://doi.org/10.1038/s41587-019-0209-9.

Cao, J., Yan, J., Zhao, Y., Jiang, W., 2013. Effects of postharvest salicylic acid dipping on *Alternaria* rot and disease resistance of jujube fruit during storage. J. Sci. Food Agric. 93, 3252–3258. https://doi.org/10.1002/jsfa.6167.

Chen, H., Cheng, Z., Wisniewski, M., Liu, Y., Liu, J., 2015. Ecofriendly hot water treatment reduces postharvest decay and elicits defense response in kiwifruit. Environ. Sci. Pollut. Res. Int. 22, 15037–15045. https://doi.org/10.1007/s11356-015-4714-1.

Choi, H.R., Baek, M.W., Jeong, C.S., Tilahun, S., 2022. Comparative transcriptome analysis of softening and ripening-related genes in kiwifruit cultivars treated with ethylene. Curr. Issues Mol. Biol. 44, 2593–2613. https://doi.org/10.3390/cimb44060177.

Dai, Y., Wang, Z., Leng, J., Sui, Y., Jiang, M., Wisniewski, M., Liu, J., Wang, Q., 2022. Eco-friendly management of postharvest fungal decays in kiwifruit. Crit. Rev. Food Sci. Nutr. 62, 8307–8318. https://doi.org/10.1080/10408398.2021.1926908.

Dukare, A.S., Paul, S., Nambi, V.E., Gupta, R.K., Singh, R., Sharma, K., Vishwakarma, R. K., 2019. Exploitation of microbial antagonists for the control of postharvest diseases of fruits: a review. Crit. Rev. Food Sci. Nutr. 59, 1498–1513. https://doi.org/10.1080/10408398.2017.1417235.

Gorordo, M.F., Lucca, M.E., Sangorrín, M.P., 2022. Biocontrol efficacy of the *Vishniacozyma Victoriae* in semi-commercial assays for the control of postharvest fungal diseases of organic pears. Curr. Microbiol. 79, 259. https://doi.org/10.1007/s00284-022-02934-1.

Lin, M., Gao, Z., Wang, X., Huo, H., Mao, J., Gong, X., Chen, L., Ma, S., Cao, Y., 2023. Eco-friendly managements and molecular mechanisms for improving postharvest quality and extending shelf life of kiwifruit: a review. Int. J. Biol. Macromol., 128450 https://doi.org/10.1016/j.ijbiomac.2023.128450.

Lin, S., Chen, C., Luo, H., Xu, W., Zhang, H., Tian, J., Ju, R., Wang, L., 2019. The combined effect of ozone treatment and polyethylene packaging on postharvest quality and biodiversity of *Toona sinensis* (A.Juss.) M.Roem. Postharvest Biol. Technol. 154, 1–10. https://doi.org/10.1016/j.postharvbio.2019.04.010.

Lippi, G., Mattiuzzi, C., 2020. Kiwifruit and cancer: an overview of biological evidence. Nutr. Cancer 72, 547–553. https://doi.org/10.1080/01635581.2019.1650190.

Liu, J., Sui, Y., Chen, H., Liu, Y., Liu, Y., 2018. Proteomic analysis of kiwifruit in response to the postharvest pathogen, *Botrytis cinerea*. Front. Plant Sci. 9, 158. https://doi.org/10.3389/fpls.2018.00158.

Liu, X., Gao, Y., Yang, H., Li, L., Jiang, Y., Li, Y., Zheng, J., 2020. *Pichia kudriavzevii* retards fungal decay by influencing the fungal community succession during cherry tomato fruit storage. Food Microbiol. 88, 103404 https://doi.org/10.1016/j.fm.2019.103404.

Ma, S., Wang, Y., Teng, W., 2023. *Bacillus velezensis* K-9 as a potential biocontrol agent for managing potato scab. Plant Dis. 107, 3943–3951. https://doi.org/10.1094/PDIS-12-22-2829-RE.

Mendes, R., Garbeva, P., Raaijmakers, J.M., 2013. The rhizosphere microbiome: significance of plant beneficial, plant pathogenic, and human pathogenic microorganisms. FEMS Microbiol. Rev. 37 (5), 634–663. https://doi.org/10.1111/1574-6976.12028.

Nian, L., Xie, Y., Zhang, H., Wang, M., Yuan, B., Cheng, S., Cao, C., 2023. *Vishniacozyma victoriae*: an endophytic antagonist yeast of kiwifruit with biocontrol effect to *Botrytis cinerea*. Food Chem. 411, 135442 https://doi.org/10.1016/j.foodchem.2023.135442.

Quast, C., Pruesse, E., Yilmaz, P., Gerken, J., Schweer, T., Yarza, P., Peplies, J., Glöckner, F.O., 2012. The silva ribosomal RNA gene database project: improved data processing and web-based tools. Nucleic Acids Res. 41, D590–D596. https://doi.org/10.1093/nar/gks1219.

Rabbee, M.F., Ali, M.S., Choi, J., Hwang, B.S., Jeong, S.C., Baek, K.H., 2019. *Bacillus velezensis*: a valuable member of bioactive molecules within plant microbiomes. Mol. (Basel, Switz.) 24 (6), 1046. https://doi.org/10.3390/molecules24061046.

Ren, X., Zhang, Q., Zhang, W., Mao, J., Li, P., 2020. Control of aflatoxigenic molds by antagonistic microorganisms: inhibitory behaviors, bioactive compounds, related mechanisms, and influencing factors. Toxins 12 (1), 24. https://doi.org/10.3390/toxins12010024.

Shi, Y., Yang, Q., Zhao, Q., Dhanasekaran, S., Ahima, J., Zhang, X., Zhou, S., Droby, S., Zhang, H., 2022b. *Aureobasidium pullulans* S-2 reduced the disease incidence of tomato by influencing the postharvest microbiome during storage. Postharvest Biol. Technol. 185, 111809 https://doi.org/10.1016/j.postharvbio.2021.111809.

Shi, Y., Yang, Q., Zhang, Q., Zhao, Q., Godana, E., Zhang, X., Zhou, S., Zhang, H., 2022a. The preharvest application of *Aureobasidium pullulans* S2 remodeled the microbiome of tomato surface and reduced postharvest disease incidence of tomato fruit. Postharvest Biol. Technol. 194, 112101 https://doi.org/10.1016/j.postharvbio.2022.112101.

Shu, P., Zhang, Z., Wu, Y., Chen, Y., Li, K., Deng, H., Zhang, J., Zhang, X., Wang, J., Liu, Z., Xie, Y., Du, K., Li, M., Bouzayen, M., Hong, Y., Zhang, Y., Liu, M., 2023. A comprehensive metabolic map reveals major quality regulations in red-flesh kiwifruit (*Actinidia chinensis*). N. Phytol. 238, 2064–2079. https://doi.org/10.1111/nph.18840.

Sun, K., Zhang, X., Wei, Z., Wang, Z., Liu, J., Liu, J., Gao, J., Guo, J., Zhao, X., 2023. Analysis of metabolic and transcription levels provides insights into the interactions of plant hormones and crosstalk with MAPKs in the early signaling response of cherry tomato fruit induced by the yeast cell wall. Food Chem. Mol. Sci. 6, 100160 https://doi.org/10.1016/j.fochms.2022.100160.

Thambugala, K.M., Daranagama, D.A., Phillips, A.J.L., Kannangara, S.D., Promputtha, I., 2020. Fungi vs. fungi in biocontrol: an overview of fungal antagonists applied against fungal plant pathogens. Front. Cell. Infect. Microbiol. 10, 604923 https://doi.org/10.3389/fcimb.2020.604923.

Vu, T.X., Tran, T.B., Tran, M.B., Do, T.T.K., Do, L.M., Dinh, M.T., Thai, H.D., Pham, D.N., Tran, V.T., 2023. Efficient control of the fungal pathogens *Colletotrichum gloeosporioides* and *Penicillium digitatum* infecting citrus fruits by native soilborne *Bacillus velezensis* strains. Heliyon 9, e13663. https://doi.org/10.1016/j.heliyon.2023.e13663.

Wang, Z., Sui, Y., Li, J., Tian, X., Wang, Q., 2022. Biological control of postharvest fungal decays in citrus: a review. Crit. Rev. Food Sci. Nutr. 62, 861–870. https://doi.org/10.1080/10408398.2020.1829542.

Wicaksono, W.A., Buko, A., Kusstatscher, P., Cernava, T., Sinkkonen, A., Laitinen, O.H., Virtanen, S.M., Hyöty, H., Berg, G., 2023. Impact of cultivation and origin on the fruit microbiome of apples and blueberries and implications for the exposome. Microb. Ecol. 86, 973–984. https://doi.org/10.1007/s00248-022-02157-8.

Yang, F., Zhao, R., Suo, J., Ding, Y., Tan, J., Zhu, Q., Ma, Y., 2024. Understanding quality differences between kiwifruit varieties during softening. Food Chem. 430, 136983 https://doi.org/10.1016/j.foodchem.2023.136983.

Zhao, Q., Shi, Y., Legrand Ngolong Ngea, G., Zhang, X., Yang, Q., Zhang, Q., Xu, X., Zhang, H., 2023. Changes of the microbial community in kiwifruit during storage after postharvest application of *Wickerhamomyces anomalus*. Food Chem. 404, 134593 https://doi.org/10.1016/j.foodchem.2022.134593.

Zhimo, V.Y., Kumar, A., Biasi, A., Salim, S., Feygenberg, O., Toamy, M.A., Abdelfattaah, A., Medina, S., Freilich, S., Wisniewski, M., Droby, S., 2021. Compositional shifts in the strawberry fruit microbiome in response to near-harvest application of *Metschnikowia fructicola*, a yeast biocontrol agent. Postharvest Biol. Technol. 175, 111469 https://doi.org/10.1016/j.postharvbio.2021.111469.